

SUPPLEMENTARY RESULTS: CONPCO

This document contains the model-specific component and loss-function ablations for ConPCO, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified $[0, 10]$ score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

II. RESULTS

ConPCO exhibits a more aspect-dependent response. BOMA is best on five CCC-BS aspects, while the Baseline remains best for utterance fluency and prosody, +MS is best for utterance accuracy, and +DB is best for utterance completeness (Table S1). BOMA achieves the lowest Macro-MSE on seven aspects, with the remaining two favoring +DB and the Baseline (Table S2). PCC is highest for the Baseline on eight aspects, with +DB leading utterance completeness (Table S3). Under the controlled loss comparison, DB-MSE is best on eight CCC-BS aspects and all nine Macro-MSE aspects (Tables S4 and S5).

TABLE S1
ABLATION STUDY FOR CONPCO. CORRELATIONS ARE MEASURED USING CCC-BS.

Configuration	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.470	0.545	0.087	0.591	0.716	0.061	0.883	0.880	0.788
+ CV	0.450	0.504	0.080	0.537	0.716	0.051	0.772	0.804	0.781
+ DB	0.590	0.635	0.523	0.622	0.656	0.376	0.804	0.811	0.766
+ MS	0.494	0.519	0.083	0.559	0.751	0.042	0.724	0.745	0.786
BOMA	0.628	0.678	0.596	0.655	0.745	0.269	0.866	0.876	0.819

^a The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR CONPCO. LOWER VALUES INDICATE BETTER PERFORMANCE.

	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
Configuration	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
Baseline	1.873	15.940	2.346	13.053	0.358	12.498	1.791	10.482	1.049	2.880	0.238	26.253	0.628	2.149	0.678	1.970	0.938	3.430
+ CV	2.001	16.026	2.579	14.429	0.449	12.880	2.002	12.035	1.087	2.928	0.244	26.764	0.719	3.765	0.750	3.042	0.997	3.461
+ DB	3.579	8.970	3.169	7.785	0.952	5.555	2.326	7.549	1.160	3.120	0.157	16.025	0.724	3.596	0.783	3.071	1.024	3.757
+ MS	2.541	13.189	2.606	12.266	0.462	12.848	2.021	10.286	1.240	2.354	0.247	27.058	0.820	4.474	0.866	3.912	1.129	3.177
BOMA	6.793	6.524	5.315	6.076	1.611	4.401	3.700	5.830	1.449	2.229	0.203	17.699	0.815	2.313	0.823	1.959	1.227	2.667

^a Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified $[0, 10]$ scale; lower values (↓) indicate better performance.

^b The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^c Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3
PCC ABLATION STUDY FOR CONPCO. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone Score	Word Score (PCC)			Utterance Score (PCC)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.701	0.669	0.437	0.682	0.780	0.749	0.830	0.823	0.803
+ CV	0.642	0.579	0.258	0.594	0.748	0.510	0.819	0.812	0.778
+ DB	0.575	0.527	0.249	0.533	0.739	0.773	0.819	0.809	0.780
+ MS	0.590	0.548	0.253	0.565	0.741	0.334	0.800	0.790	0.767
BOMA	0.519	0.472	0.201	0.478	0.723	0.647	0.808	0.804	0.764

^a The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR CONPCO. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
MSE	0.494	0.519	0.083	0.559	0.751	0.042	0.724	0.745	0.786
DB-MSE	0.628	0.678	0.596	0.655	0.745	0.269	0.866	0.876	0.819
SB	0.524	0.530	0.084	0.570	0.750	0.040	0.720	0.734	0.796

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR CONPCO. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓		
MSE	2.541	13.189	2.606	12.266	0.462	12.848	2.021	10.286	1.240	2.354	0.247	27.058	0.820	4.474	0.866	3.912	1.129	3.177
DB-MSE	6.793	6.524	5.315	6.076	1.611	4.401	3.700	5.830	1.449	2.229	0.203	17.699	0.815	2.313	0.823	1.959	1.227	2.667
SB	2.733	12.023	2.688	11.525	0.454	12.737	2.074	9.707	1.438	2.284	0.248	26.940	0.924	4.455	0.917	3.979	1.342	2.913

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

^e Bold values indicate the lowest error for each metric within the model-specific loss ablation.